

1.Experiment: FIND-S Algorithm

To implement and demonstrate the **FIND-S algorithm** for finding the **most specific hypothesis** using training data.

Python Program

```
import pandas as pd
from google.colab import files
uploaded = files.upload()
filename = list(uploaded.keys())[0]
df = pd.read_csv(filename)
print("Training Data:\n")
print(df)
hypothesis = None
for i in range(len(df)):
    if df.iloc[i, -1] == "Yes":
        if hypothesis is None:
            hypothesis = list(df.iloc[i, :-1])
        else:
            for j in range(len(hypothesis)):
                if hypothesis[j] != df.iloc[i, j]:
                    hypothesis[j] = "?"
```

```
print("\nFinal Most Specific Hypothesis:")
```

```
print(hypothesis)
```

Output

```
Final Most Specific Hypothesis:  
['?', '?', '?', '?', '?', 'Same', '?', 'Outdoor', '?']
```